

Unrelated Is Not Opposed: A Signed Scale for Meaning Preservation

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Abstract

Metrics for meaning preservation report how much meaning two sentences share, on a scale from 0 to 100. That scale cannot express opposition. Because contradicting sentences are about the same thing, they overlap heavily, and a released model scores “I drink milk” against “I do not drink milk” at 58 out of 100. We propose a signed scale on $[-100, 100]$, obtained by multiplying a magnitude head by $1 - \alpha p_{\text{contra}}$ from a three-way polarity head. The form is a product so that unrelated sentences score 0 whatever their polarity. Thresholded token overlap ranks polarity better than the released metric does, 63.5% AUC against 60.8%. Over seven encoders and ten seeds a polarity head reaches 91.1% macro- F_1 , derived-polarity augmentation leaves task performance unchanged while raising generated sanity suites from 79–90% to 98–99% on every one of them, and composition puts 83.7% of SICK contradictions below zero where the magnitude alone puts none, cutting from 14.1% to 2.1% the contradictions a system would accept as preserved meaning.

1 Introduction

Assessing whether a rewritten sentence preserves the meaning of its source is a prerequisite for evaluating text simplification, summarisation and translation (Saggion, 2017). The metrics built for it, whether embedding-based (Zhang et al., 2020; Reimers and Gurevych, 2019), question-based (Scialom et al., 2021) or trained end to end (Maddela et al., 2023; Beauchemin et al., 2023), all return a single non-negative number, where 0 means the sentences have nothing in common and 100 means the meaning is intact.

That output shape leaves no room for opposition. Two sentences that contradict each other are, by construction, about the same thing, so they share most of their lexical material and most of their subject matter, and a magnitude metric reads that overlap as preserved meaning. A released model scores “I drink milk” against “I do not drink milk” at 58.3, and the SICK contradiction “two dogs are wrestling and hugging” against “there is no dog wrestling and hugging” at 51.7. Both

land on the positive half of a scale whose positive half means the meaning survived. Baral (2026) quantified the same effect on legal text, where minimal perturbations that reverse legal force while preserving most tokens move embedding metrics over a small fraction of their range. These metrics are not imprecise on such pairs; they are confidently wrong in the direction that matters.

The failure is measurable on a released metric. Evaluated on a held-out set of 12 000 class-balanced pairs, the published MeaningBERT model ranks agreement above contradiction with an AUC of 60.8%, and on SICK (Marelli et al., 2014) alone it reaches 48.0%, scoring contradictions marginally higher than agreements. A thresholded count of shared tokens reaches 63.5% on the same split. A metric trained on human meaning-preservation judgements therefore orders polarity no better than lexical overlap.

This paper builds a metric on a signed scale, $[-100, 100]$, where -100 denotes opposed meaning and 0 denotes no relation. The diagnosis above is the motivation; what we contribute is the model that answers it. Our contributions are:

1. **A meaning-preservation metric that can be negative** (Section 3, Section 7). To our knowledge the first that distinguishes “these sentences say the opposite” from “these sentences share little”, rather than collapsing both onto a low non-negative score. The metric multiplies a magnitude head by a polarity term, and its one free parameter is fixed by requiring the negative half to be reachable rather than fitted against the quantity it is meant to produce. At a fixed acceptance threshold it cuts the contradictions wrongly accepted as preserved meaning from 14.10% to 2.13%, for seven tenths of a point of entailment recall, and it trains nothing beyond a three-way classifier.
2. **The polarity head the metric needs, and the corpus it is trained on** (Section 4, Section 6). A merged corpus stratified by class and by corpus with source-sentence leakage removed in both

directions, plus an augmentation whose polarity is derived rather than copied, since mirroring a contradiction preserves its label and mirroring an entailment does not. Over a grid of seven encoders, ten seeds and two corpus conditions, the head reaches 91.1% macro-F₁ and the augmentation repairs behaviour at the ends of the scale on every encoder, which is where a signed score is read.

2 Related Work

2.1 Meaning Preservation Metrics

BERTScore (Zhang et al., 2020) computes token-level similarity from contextual embeddings, SARI (Xu et al., 2016) scores edit operations, QuestEval (Scialom et al., 2021) asks questions of both texts, and LENS (Maddela et al., 2023) and MeaningBERT (Beauchemin et al., 2023) are trained against human judgements. All emit a magnitude, and all are evaluated by correlation with human ratings plus, for MeaningBERT, two sanity checks: identical sentences must score 100 and unrelated sentences 0. Baral (2026) observes that both checks are satisfied by any monotone function of lexical overlap, which is why they do not catch the failure studied here.

2.2 Specialised Metrics

A second line of work argues that a general-purpose metric is the wrong instrument where a small edit carries a large consequence. Beauchemin et al. (2025) build a metric for legal meaning preservation, on the ground that preservation in a specialised domain differs from preservation in ordinary text and that general metrics fail the same two sanity checks their own passes. Lyu and Pergola (2024) make the parallel argument for biomedical simplification, where surface-level metrics do not register whether the core content survived. Both specialise the domain while keeping a non-negative output. Our proposal is orthogonal: it changes the shape of the scale rather than the domain it is fitted on, and the two compose, since a domain-specific magnitude head can be substituted into Equation 1 without touching the polarity half.

2.3 Inference and Its Held-Out Suites

Natural language inference supplies the missing dimension. Large-scale corpora (Williams et al., 2018; Nie et al., 2020) and fact verification (Thorne et al., 2018; Schuster et al., 2021) provide three-way entailment, neutral and contradiction labels, while targeted suites test what the aggregate hides: Geiger et al. (2020) probes downward monotonicity, where a lexical sub-

Pair	m	p_{contra} (%)	s
“I drink milk” / “I consume milk”	96.70	0.50	+95.80
“I drink milk” / “I do not drink milk”	58.30	99.70	−58.00
“I drink milk” / “the meeting is at three”	0.00	0.50	+0.00
“Two dogs are wrestling” / “there is no dog wrestling”	51.70	99.40	−51.10

Table 1: Measured, not illustrative. m from a released magnitude model, p_{contra} from our polarity head, s from Equation 1 at $\alpha = 2$. Rows two and four are the cases a magnitude scale gets backwards.

stitution inside the scope of a negation reverses the direction of entailment, and NaN-NLI (Truong et al., 2022) probes negation scoping below the clause. An inference model says whether the second sentence follows from or denies the first and nothing about how much meaning the two share; a preservation metric answers the converse. Marelli et al. (2014) is, to our knowledge, the only resource annotating both over the same pairs, which makes it the only place the composition can be calibrated.

3 A Signed Scale

We define the score so that +100 is identical meaning, 0 is no relation and −100 is opposed meaning. The decisive property is that the negative half measures active contradiction rather than the absence of shared meaning: two sentences about different topics score 0, not −100, since to contradict each other they must first be about the same thing. The score therefore carries two quantities. The magnitude asks what the sentences talk about together and is exactly what existing metrics estimate; the polarity asks, once the subject is shared, whether they assert the same thing or its negation.

We compose the two as

$$s = \text{clip}_{[-100,100]}(m \times (1 - \alpha p_{\text{contra}})), \quad (1)$$

with $m \in [0, 100]$ from a magnitude head and $p_{\text{contra}} \in [0, 1]$ from a three-way polarity head. The form is a product and not a difference: an unrelated pair has $m \approx 0$ and scores near zero whatever its polarity, whereas a subtraction would let it drift negative and destroy the distinction the scale rests on. Table 1 gives four measured cases and Figure 1 traces one of them through both heads.

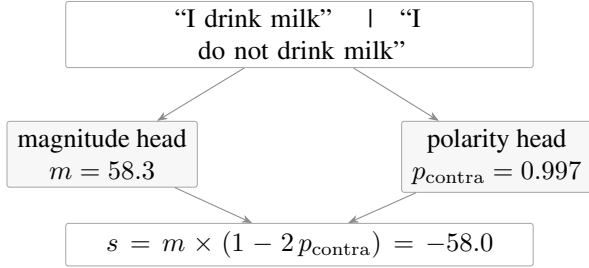


Figure 1: One pair, end to end. The two heads are read independently and Equation 1 turns a middling positive score into a negative one.

The slope α governs how much contradiction is needed to flip the sign. A certain contradiction on a perfect magnitude lands at $100 \times (1 - \alpha)$, so the negative half is reachable only from $\alpha = 2$ upwards, and below it the scale is truncated. However, we treat reachability as a constraint on the search rather than as a term in the objective, since it defines the scale rather than trading against its other properties, and Section 7 reports what the constraint costs.

Two alternative designs were rejected. Typed penalties in the style of MQM, subtracting a cost per error category from a perfect score, give an interpretable and natively negative output but require an error taxonomy annotated over the same pairs, and they replace the target rather than extend it. Learning $[-100, 100]$ end to end requires manufacturing the target from categorical labels, which amounts to training a model on a conversion rule and then reporting the rule. Composition avoids both problems and keeps each head separately falsifiable: when the composed score is wrong, the responsible half is identifiable.

4 Data

4.1 What Each Corpus Contributes

The polarity head needs volume, a bridge between the two dimensions it composes, and a way of seeing the failures an aggregate score hides. No single corpus provides all three, so we merge four (Table 2), each chosen for one of them.

VitaminC (Schuster et al., 2021) supplies the volume, and supplies it in the regime the paper is about. Its 488 904 pairs are contrastive by construction: a Wikipedia sentence was edited so that the claim its previous revision supported is refuted by the next, which produces contradictions differing from their source by a few tokens rather than by topic. No other corpus has that at scale. SICK (Marelli et al., 2014) supplies the bridge, since it carries a relatedness score and an inference label over the same pairs, and the composition

Corpus	Rows	Labels	Use
VitaminC	488 904	FACT3	split
SICK	9 840	NLI3 + rel.	split
MoNLI	1 202	ent./neutral	held-out
NaN-NLI	258	NLI3	held-out

Table 2: Corpora. FACT3 is SUPPORTS, NOT ENOUGH INFO, REFUTES, mapped onto NLI3. Split: merged, then divided into train, development and test. SICK alone carries a relatedness score, which the composition is calibrated against.

needs both at once: its third criterion asks that the score still track human relatedness where the sign does not flip. SICK is therefore the only corpus on which α can be calibrated at all. MoNLI (Geiger et al., 2020) and NaN-NLI (Truong et al., 2022) supply the view of the failures, being small and adversarial by design. They never enter training and serve as held-out sets, because a head that scores well on VitaminC while failing them has learned the corpus and not the relation.

We considered and rejected PAWS (Zhang et al., 2019), whose high-overlap non-paraphrases look like the right material. Its binary label is not a polarity judgement: two sentences that are not paraphrases need not contradict each other, and mapping “not a paraphrase” onto contradiction would teach the head that difference amounts to opposition, which is the failure this work exists to remove.

4.2 Augmenting the Ends of the Scale

Natural corpora are thin exactly where a signed score is read: they contain few identical pairs and few pairs with nothing in common, because neither is an interesting inference problem. A second corpus condition therefore adds three generated families in equal quotas of 25 000, one per behaviour the scale depends on. Identical pairs, labelled entailment, anchor the top of the scale; low-overlap random pairs, labelled neutral, anchor the middle, and this is the family that teaches that unrelated is not opposed; mirrored contradictions teach that opposition is symmetric.

Every generated label is derived and never copied, because polarity is not symmetric the way magnitude is. If A contradicts B then B contradicts A , so a mirrored contradiction stays one; but “a dog is running” entails “an animal is running” while the converse is false, so a mirrored entailment loses its label rather than carrying a wrong one. The quotas are equal for the same reason: mirroring alone would take the contradiction class from 50 641 to 101 282 against 51 274 entailments, and bias the head towards the one class whose sign inverts the

score.

4.3 Splitting a Merged Corpus of Unequal Parts

Two properties are wanted from the development and test splits, balance and the absence of leakage, and a naive split gives neither.

Drawing at random leaves the split at 8% SICK, because VitaminC is fifty times its size, so a model selected on it is selected almost entirely on Wikipedia claim-evidence pairs. We therefore draw 4 000 rows per class in equal share from each corpus, taking a corpus whole when it is smaller than its share and redistributing the shortfall, which raises SICK to 34% of the test split. Leakage is the second problem and runs in both directions. VitaminC reuses the same evidence sentence across many claims and splits by case rather than by evidence, so a random split puts the same source on both sides of the wall: 5.6% of our test groups shared a source sentence with training. We therefore group rows by source sentence and move whole groups, then assert that no pair and no source sentence crosses between train, development and test, before and after augmentation. Appendix A gives the construction and the exact counts.

5 Experimental Setup

We train the polarity head on seven encoders, each on both corpus conditions, and repeat every cell over ten seeds so that no claim rests on a single draw. The encoders span three families, BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019) and DeBERTaV3 (He et al., 2023), in two sizes, and include both a plain and an inference-pretrained variant of the same architecture, which is what lets the effect of inference pretraining be read at constant size. The two conditions are RAW, the merged corpus at 152 060 training rows, and AUG, the same corpus plus the generated families at 227 060. Checkpoints already fine-tuned for three-way inference keep their head when its class order matches ours and have their classifier rows permuted when the published order is reversed. All runs share one set of optimisation settings, given with the hardware in Appendix B and Appendix C.

We report macro- F_1 on the stratified test split, the AUC of $1 - p_{\text{contra}}$ ranking entailment above contradiction, accuracy on three generated sanity suites built from test-split sentences, and accuracy on the two held-out sets. Significance uses Welch’s t -test (Welch, 1947) between conditions and a paired test by seed between architectures, with Cohen’s d beside every p -value, since ten seeds with standard deviations near two tenths of a point make a difference of one tenth significant and

meaningless at once. Holm-Bonferroni (Holm, 1979) across the fourteen condition tests changes no verdict, so the p -values below are uncorrected.

6 Results

The grid answers three questions: whether the polarity the signed scale needs is learnable at all, what the augmentation changes, and whether meaning-preservation data could have supplied the same signal. The best configuration reaches $91.1 \pm 0.14\%$ macro- F_1 on a class-balanced test set of 12 000 pairs (Table 3), with a seed standard deviation of 0.14 points over ten runs. Three references bound it from below, reported as macro- F_1 in the caption of Table 6. The majority class sits at the arithmetic floor of 16.7%, Jaccard overlap between the two token sets cut by two thresholds fitted on development data reaches 39.4%, and a TF-IDF bag of unigrams and bigrams with a linear model reaches 45.6%. Consequently, between the floor and the fine-tuned head, token overlap covers 30.5% of the achievable range and TF-IDF 38.9%, so roughly two fifths of the task is surface statistics of the corpus and the head is worth 45 points above the best of the three.

Augmentation then buys sanity rather than performance. On the four DeBERTa encoders the macro- F_1 difference between conditions lies between -0.21 and $+0.08$ points and is not significant (p from 0.17 to 0.74); on the other three it is significant but below one point. The sanity suites move in a different regime (Table 4, Figure 2): $+8.4$ to $+20.0$ points, $p < 10^{-4}$ on all seven encoders, with d from 9.9 to 81.1. The effect sizes settle the matter: $d = 81$ on RoBERTa-base_{STS-B} sits two orders of magnitude above the seed noise, whereas the largest task effect in the grid is $d = 2.2$ for 0.73 points. Every encoder converges to 98.3–98.8% regardless of where it started, and the clearest cases are the weakest models: RoBERTa-base_{STS-B} moves from 78.7% to 98.7% and BERT-base from 79.4% to 98.3%. Thus, the augmentation does not make the head better at the task; it repairs how the head behaves at the ends of the scale, which is where a signed metric is read.

The sanity suites come from the same three generators as the augmented training rows, over different sentences but by the same construction, so an AUG model is partly evaluated on a shape it has seen and the gain is not a claim of generalisation to arbitrary degenerate pairs. What it establishes is that the behaviour is learnable and not acquired incidentally: seven encoders trained without these rows converge nowhere near 98%.

The gain is concentrated on two of the three families. Table 5 separates the suites, and identical pairs

Encoder	Condition	Macro-F ₁	Sanity suites	NaN-NLI	MoNLI
DeBERTa-v3-large _{NLI}	RAW	91.10 ± 0.14	90.35 ± 0.47	72.95 ± 0.65	34.93 ± 3.13
	AUG	90.98 ± 0.31	98.76 ± 0.04	71.90 ± 1.83	31.23 ± 5.11
DeBERTa-v3-large	RAW	90.91 ± 0.10	88.89 ± 0.42	70.62 ± 1.17	22.79 ± 5.22
	AUG	90.70 ± 0.43	98.70 ± 0.40	69.11 ± 1.24	15.97 ± 5.84
RoBERTa-large _{MNLI}	RAW	89.52 ± 0.10	84.59 ± 1.07	68.06 ± 1.02	7.02 ± 1.10
	AUG	89.70 ± 0.10	98.69 ± 0.07	66.59 ± 1.09	8.34 ± 1.68
DeBERTa-v3-base _{NLI}	RAW	89.48 ± 0.15	87.46 ± 0.37	68.57 ± 0.62	11.47 ± 2.39
	AUG	89.56 ± 0.17	98.74 ± 0.08	67.40 ± 0.67	11.62 ± 2.05
DeBERTa-v3-base	RAW	88.81 ± 0.63	86.32 ± 1.74	66.40 ± 1.68	13.42 ± 2.96
	AUG	88.73 ± 0.35	98.60 ± 0.13	64.11 ± 1.92	10.87 ± 3.27
RoBERTa-base _{STS-B}	RAW	85.91 ± 0.15	78.67 ± 0.34	63.76 ± 0.99	4.60 ± 0.60
	AUG	86.22 ± 0.13	98.65 ± 0.08	62.60 ± 0.74	4.94 ± 0.64
BERT-base	RAW	77.72 ± 0.38	79.38 ± 0.65	46.36 ± 2.70	13.64 ± 5.52
	AUG	78.45 ± 0.26	98.31 ± 0.10	49.03 ± 1.13	17.48 ± 2.57

Table 3: Polarity head, mean and standard deviation over ten seeds, in percent. AUG adds the derived-polarity augmentation to RAW. NaN-NLI and MoNLI are held-out sets. **Bold**: best per column.

Encoder	Macro-F ₁		Sanity suites	
	Δ (pp)	d	Δ (pp)	d
DeBERTa-v3-large _{NLI}	-0.11 ± 0.11 [†]	-0.47	+8.40 ± 0.15	+25.4
DeBERTa-v3-large	-0.21 ± 0.14 [†]	-0.67	+9.81 ± 0.18	+24.1
RoBERTa-large _{MNLI}	+0.19 ± 0.05	+1.79	+14.10 ± 0.34	+18.6
DeBERTa-v3-base _{NLI}	+0.08 ± 0.07 [†]	+0.49	+11.28 ± 0.12	+42.2
DeBERTa-v3-base	-0.08 ± 0.23 [†]	-0.15	+12.28 ± 0.55	+9.9
RoBERTa-base _{STS-B}	+0.31 ± 0.06	+2.22	+19.98 ± 0.11	+81.1
BERT-base	+0.73 ± 0.15	+2.22	+18.94 ± 0.21	+40.9

Table 4: Effect of augmentation in points, AUG minus RAW, with the standard error of the difference and Cohen’s d (Welch, ten seeds). [†]: not significant at $p < 0.05$. Every sanity difference has $p < 10^{-4}$.

are essentially solved before augmentation. Unrelated pairs rise from 92–96% to 99.3–99.7% on the three encoders shown, and across the seven the confusion matrices put between 2.4% and 7.9% of pairs with nothing in common in the contradiction class before augmentation, which on the signed scale returns a negative score where the truth is zero. Finally, mirrored contradictions rise from 74–75% to 95.6–96.5%, and on BERT-base from 50%, so the symmetry of contradiction is not a property these encoders hold on their own.

Inference pretraining buys something the aggregate does not show. Paired by seed, DeBERTa-v3-large_{NLI} beats its plain counterpart by +0.19 points of macro-F₁ ($p = 0.002$, $d = 1.39$) and by +12.1 points on MoNLI ($p < 10^{-4}$, $d = 2.58$). At base size the pattern inverts: the task gain is larger (+0.67 points, $p = 0.005$) and the MoNLI difference is not significant. Consequently,

inference pretraining supplies a capability the aggregate test set does not exercise, which is the argument for reporting held-out suites alongside it. MoNLI accuracy is low everywhere, and the confusion matrix locates the failure: the heads almost never call these pairs contradictions, with 0.2% of true entailments predicted contradiction on DeBERTa-v3-large_{NLI} against 93% predicted neutral. They have not failed to notice the negation; they have failed to reverse the direction of entailment under it.

The headline also transfers across corpora. Stratification leaves the test split one third SICK and two thirds VitaminC. Scored separately, DeBERTa-v3-large_{NLI} reaches 91.6% macro-F₁ on SICK against 90.4% on VitaminC, and 99.3% against 98.4% AUC. These four figures and the AUC of Table 6 come from a single run, since the grid stores predicted classes rather than the probabilities an AUC needs, and they carry no seed

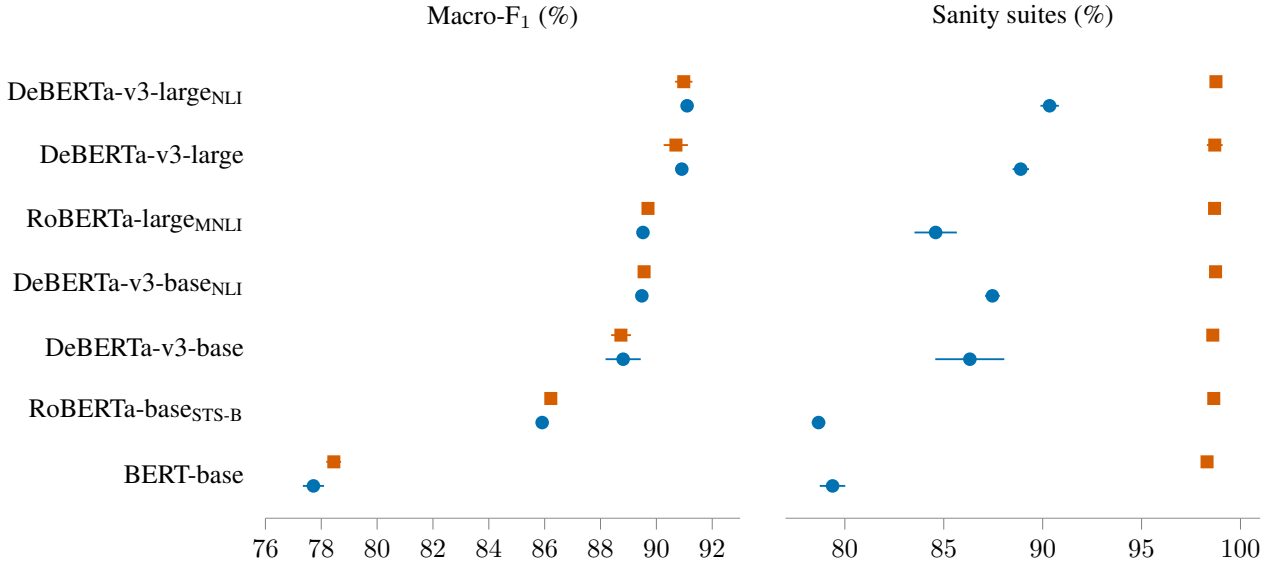


Figure 2: Augmentation moves one axis and not the other. **RAW** and **AUG**: means over ten seeds, bars one standard deviation. The panels do not share an x range.

Encoder	Condition	Identical	Unrelated	Mirrored
DeBERTa-v3-large _{NLI}	RAW	100.00 $\pm$ 0.00	95.91 $\pm$ 0.78	75.14 $\pm$ 1.20
	AUG	100.00 $\pm$ 0.00	99.73 $\pm$ 0.07	96.54 $\pm$ 0.13
DeBERTa-v3-large	RAW	100.00 $\pm$ 0.00	92.75 $\pm$ 0.60	73.93 $\pm$ 1.46
	AUG	99.99 $\pm$ 0.04	99.71 $\pm$ 0.13	96.41 $\pm$ 1.14
BERT-base	RAW	96.46 $\pm$ 0.16	91.99 $\pm$ 0.67	49.68 $\pm$ 1.74
	AUG	100.00 $\pm$ 0.00	99.33 $\pm$ 0.09	95.61 $\pm$ 0.29

Table 5: The three generated suites separately, ten seeds, in percent, for the best encoder, its plain counterpart and the weakest of the grid. Identical pairs must be entailment, unrelated pairs neutral, mirrored pairs contradiction.

variance. The aggregate is therefore not an artefact of the larger corpus.

Polarity, finally, does not follow from preservation data at all. Table 6 scores three released magnitude models on both tasks at once, with the polarity heads and the surface baselines beneath them for reference. The two columns do not order the models the same way: the weakest magnitude model, at $r = 0.323$, ranks polarity better than one with twice its correlation, and that second model, a competent magnitude estimator at $r = 0.640$, sits near chance at 53.7%. Indeed, nothing in a magnitude metric’s training signal distinguishes a contradiction from a failed paraphrase, since both receive the same low human rating, so no quantity of meaning-preservation data can produce polarity. The exception is instructive: DeBERTa-v3-large reaches 81.5% without a single polarity label, while a BERT-base model trained on the same corpus reaches 53.7%, which places the separation in the encoder’s pretraining rather than in the preservation data.

An off-the-shelf inference model used as published ranks entailment above contradiction at 96.6%, above every magnitude model, which bounds what fine-tuning contributes. Fine-tuning the same architecture adds +22.7 points of macro-F₁, and on the single run where both are measured it adds +2.1 points of AUC, so it improves both the ordering of p_{contra} and its calibration into three classes, but the ordering was largely available without it. The macro-F₁ gap flatters fine-tuning, since the off-the-shelf head never saw our label mapping; the AUC gap does not. Finally, a practitioner who needs only the sign may reach most of the way with no training at all.

7 Composing the Two Heads

7.1 Fitting the Slope

We fit α on the SICK half of the development split and report on the SICK half of the test split, 4 116 pairs of which 712 are contradictions. The components are a released magnitude model and our best polarity

Model	Preserv. r	Polarity AUC
<i>Magnitude metrics</i>		
MeaningBERT (published)	0.323	60.80
BERT-base, retrained	0.640	53.70
DeBERTa-v3-large, retrained	0.705	81.50
<i>Polarity heads, published then ours</i>		
DeBERTa-v3-large _{MNLI}	–	96.62
DeBERTa-v3-large _{NLI}	–	98.72
<i>Surface baselines</i>		
Token overlap, Jaccard	–	63.50
TF-IDF, linear model	–	68.50

Table 6: Both tasks on one test split. r is on the 1 536 human-annotated preservation pairs, excluding 116 generated controls that inflate it by up to +0.21; AUC ranks entailment above contradiction on our 12 000 pairs. Dashes: no preservation score. Three-way macro- F_1 of the baselines, 16.67, 39.36 and 45.64, and 68.36 for the off-the-shelf head, against 91.10 for the fine-tuned one.

head, DeBERTa-v3-large_{NLI}. Since no signed gold score exists, α is selected against three requirements multiplied together, so that sacrificing one cannot hide behind the others:

$$J(\alpha) = c(\alpha) \cdot e(\alpha) \cdot \max(r(\alpha), 0), \quad (2)$$

where c is the share of contradictions scoring below zero, e the share of entailments scoring above it, and r the Pearson correlation between the composed score and human relatedness on the pairs whose sign does not flip. A product and not a mean: a sum would let a composition trade the sign away for correlation and still report a respectable number, and a composition sending every contradiction to -100 and everything else to noise maximises the first two factors and is excluded by the third. The development split selects $\alpha = 2$, the value the reachability constraint of Section 3 also forces.

Table 7 gives the result and Figure 3 shows it. The neutral column is where the composition is slightly worse than what it extends: 97.45% of neutral pairs stay strictly positive against 98.35% for the magnitude alone. SICK neutrals are related captions, not the unrelated pairs of the title, so this column shows the product leaving alone what it has no reason to touch; the title’s property is tested in Table 5. Nine tenths of a point is the cost of making the scale signed for the class that should never cross zero, and it is a cost rather than a gain because the product can only move a score down-

Slope	Contra. < 0	Neutral > 0	Rel. r	Floor
$\alpha = 1.00$	0.00	98.35	0.833	0
$\alpha = 1.25$	81.18	97.75	0.825	−25
$\alpha = 1.50$	82.72	97.65	0.816	−50
$\alpha = 1.75$	83.29	97.50	0.806	−75
$\alpha = 2.00$	83.71	97.45	0.795	−100
$\alpha = 3.00$	84.69	96.90	0.745	−100
Magnitude alone	0.00	98.35	0.852	0

Table 7: Composition on the SICK test half, in percent, over 1 404 entailments, 2 000 neutral pairs and 712 contradictions. The slope is fitted on development data, which selects $\alpha = 2$ (**bold**). Neutral: pairs left strictly positive, the class the product form exists to protect. Entailments stay positive throughout, at 100.00 everywhere except 99.93 at $\alpha = 3$. Floor: the most negative score the scale can reach.

wards. The magnitude alone puts no contradiction below the line, which is a property of a non-negative scale rather than a weakness of that model, while composition puts 83.7% of them where they belong and retains $r = 0.795$ with human relatedness on the unflipped pairs, against 0.852. The cost of requiring the full scale is visible and small: $\alpha = 1.5$ scores 1.5% higher on the combined objective and truncates the negative half at -50 .

7.2 Robustness to the Magnitude Model

Two of the three factors in Equation 2 are invariant to the magnitude head by construction. With $m \geq 0$, the sign of the product in Equation 1 is decided by p_{contra} alone, so c and e depend on the polarity head and not on the magnitude one; the magnitude sets only the amplitude. Only r can move, and the check is therefore about that term. Repeating the fit on the development split with two released checkpoints trained on the same preservation corpus, whose correlations are 0.323 and 0.705, both select $\alpha = 2$ under the reachability constraint and $\alpha = 1.5$ without it, and both send the same 84.6% of contradictions below zero, as the invariance predicts. The slope is thus stable against the quality of the magnitude head, which is a narrower statement than stability against a structurally different estimator, and that remains untested.

7.3 What Fine-Tuning the Polarity Head Buys

Changing the polarity head, in contrast, moves everything. Composing the same magnitude model with the off-the-shelf polarity head inverts the trade-off (Table 10). The off-the-shelf head is the better sign detec-

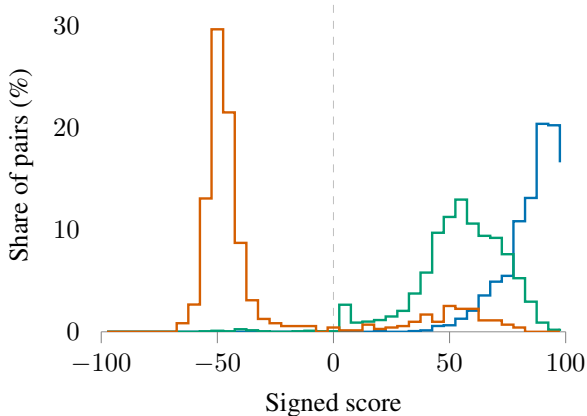


Figure 3: Signed score by gold class on the SICK test half at $\alpha = 2$, bins of width 5: **entailment** ($n = 1404$), **neutral** ($n = 2000$), **contradiction** ($n = 712$). The dashed vertical line marks zero, where the sign changes.

tor, putting 97.6% of contradictions below zero against 83.7%, but it costs 0.196 of correlation with human relatedness and loses the combined objective by 14% relative. Figure 4 shows why (Appendix D): among the pairs to which it assigns a contradiction probability near 0.5, fewer than one in ten is a contradiction, so the product drags down the magnitude of pairs that are merely unrelated. The third requirement is what makes this visible, and it is the reason α is selected against a product rather than against the sign criterion alone.

7.4 What the Sign Buys at the Decision Point

Expressing a distinction is not the same as being a better metric, and Table 8 settles the second by asking what each scale lets through. A system that accepts a rewriting when the score exceeds a cut accepts, with the magnitude alone at a cut of 50, a set in which 14.10% of the pairs are contradictions; the same cut on the signed scale leaves 2.13%. The reduction is close to sevenfold, measured against SICK’s inference labels rather than the conversion rule under test, and the bootstrap intervals do not overlap. The second block is the control, since a scale shifting everything downwards would also accept fewer contradictions: recall falls only from 99.15% to 98.43%. The advantage does not depend on where the cut falls. Sweeping it from 0 to 95 in steps of five, the signed scale never accepts a larger share of contradictions than the magnitude, and the entailment recall it gives up never exceeds 1.85 points. At a cut of 70 the magnitude already admits almost no contradiction and the scales converge, so the benefit lives where a simplification system operates rather than at the top of the range.

Scale	$s > 25$	$s > 50$	$s > 70$
<i>Contradictions accepted</i>			
Magnitude alone	17.83	14.10	0.91
Signed, published	0.49	0.15	0.00
Signed, fine-tuned	3.11	2.13	0.51
<i>Entailments kept</i>			
Magnitude alone	100.00	99.15	88.32
Signed, published	97.86	96.72	86.18
Signed, fine-tuned	99.86	98.43	86.61

Table 8: What each scale accepts as preserved meaning on the SICK test half, in percent, at three cuts. The first block is the error this paper is about; the second is the control, since a scale that merely shifted everything down would also accept fewer contradictions. Percentile intervals from 1 000 bootstrap resamplings of the pairs are at most ± 1.2 wide and do not overlap between the magnitude and either signed scale at the first two cuts.

The second row of each block separates the scale from the model that feeds it. Composing the same magnitude with an off-the-shelf inference head, used as published, accepts 0.15% contradictions at the same cut for 96.72% of entailments kept. The advantage therefore follows from the shape of the scale rather than from fine-tuning, which buys the recall the fine-tuned head keeps and the published one spends.

7.5 The Scale on Simplification Output

SICK and VitaminC are inference corpora; the metric is meant for the output of a simplification system. Scoring the 407 human-annotated pairs of the preservation test set with the composed scale sends 4.2% of them below zero, and those pairs carry a mean human rating of 38.3 against 71.0 for the rest. The sign selects the simplifications humans rated worst. On a corpus that contains no deliberate contradictions, the sign costs 0.06 of correlation with the human rating, 0.637 against 0.697 for the magnitude alone.

8 Conclusion and Future Works

A meaning-preservation metric that cannot tell a paraphrase from a negation is not imprecise but confidently wrong where precision matters, and the deployed metric has exactly this property. Making the scale signed does not require a better magnitude model; it requires a second head and a rule for combining them. The combination trains nothing beyond a three-way classifier, fits one parameter, and moves 83.7% of contradictions to the correct side of zero where the magnitude alone moves none, and what it lacks is a signed gold standard.

Limitations

While we believe the signed scale is a useful contribution to the evaluation of meaning preservation, we acknowledge seven limitations that bound our claims and offer avenues for future research.

First, there is no signed gold standard. The composition is calibrated against three necessary conditions rather than against human signed judgements, because none exist. A corpus of human-rated signed pairs would let the form of Equation 1 be tested rather than argued, and collecting one is the natural next step.

Second, the augmentation costs monotonicity. The gain in sanity is paid for on MoNLI, which drops on three of seven encoders and on both large DeBERTa variants, by -6.8 points on the plain one and -3.7 on the inference-pretrained one. Generated identical and unrelated pairs teach the easy ends of the scale, not the reversal of entailment under negation.

Third, a single corpus carries the composition. Every measurement of the signed scale lives on the SICK half of the test split, 4 116 pairs of which 712 are contradictions, which is also where α is fitted. The constraint is structural rather than a choice: SICK is the only corpus we know of annotating relatedness and inference over the same pairs, and the third criterion needs both. It binds the calibration unavoidably and the evaluation only by convenience, since Table 8 needs inference labels alone and could in principle be measured elsewhere. The split is also thin on the development side: SICK supplies 495 development rows, of which 229 shared source sentences with its own test portion and were removed, leaving the development split at 2.3% SICK against 34% in test. Model selection is therefore driven by VitaminC while the slope is fitted on SICK alone, and the two numbers are the first thing to watch should they diverge.

Fourth, the sign costs correlation. On both human-rated quantities the composition correlates worse than the magnitude alone: 0.795 against 0.852 on SICK relatedness, and 0.637 against 0.697 on preservation ratings. This is what should happen, since neither rating scale is signed and a correct negative score can only move a pair away from a non-negative target. Correlation with a non-negative rating is therefore the wrong instrument here, and Table 8 measures the benefit where it exists instead. What no existing human judgement can settle is whether the magnitude of a negative score is right, which a signed annotation would.

Fifth, the sign rests on one head. Because the composition is a product with $m \geq 0$, the sign of the score is decided entirely by p_{contra} . A polarity head that is

wrong about a pair is not corrected by a good magnitude model, and the 16.3% of SICK contradictions that stay positive are exactly the pairs the head misses.

Sixth, the property the title names is never measured on natural unrelated pairs. That two sentences about different topics score near zero follows from the product form, since $m \approx 0$ leaves the polarity term nothing to act on, and it is measured on the generated unrelated suite. No corpus we know of annotates genuinely unrelated pairs for meaning preservation, because such a pair is not an interesting annotation, so the construction carries this half of the claim and the evidence for it is self-generated.

Finally, the work is English and sentence-level throughout. Every corpus we merge is English, and the scale is defined over sentence pairs. Neither restriction is incidental. The polarity half rests on inference corpora, which are abundant in English and thin in most other languages, so the composition cannot be ported by translating a metric alone. Contradiction between documents is also a different relation from contradiction between sentences, since a document can deny another in one passage and agree with it everywhere else, and a single scalar cannot carry that. Whether the product form survives either move is untested.

Ethical Considerations

A meaning-preservation metric is deployed to decide whether a rewritten sentence may be shown to a reader in place of the original. When such a metric reports that a negated sentence preserves the meaning of its source, as we measure released models doing, the error reaches the reader rather than the engineer. The risk is concrete in the settings where simplification is most useful, where a layperson may rely on the rewritten text without access to the original (Beauchemin et al., 2023). A signed scale does not remove that risk, but it makes the failure visible to the system consuming the score instead of hiding it inside a positive number. Our corpora also inherit the biases of the Wikipedia revisions and caption corpora they are drawn from, and a polarity head trained on them carries those biases into whatever it gates (Bender et al., 2021).

Availability

The corpus builder, the loaders, the training and composition code, the result files of every cell and the script generating every table and figure in this paper will be released under an open licence. Repository: removed for double-blind review. The merged corpus inherits the most restrictive licence among its sources and will be redistributed under CC BY-NC-SA 4.0. The load-

ers are released separately so the corpus can be rebuilt from the original sources.

Use of Large Language Models

This work utilized Large Language Models (specifically Claude) to assist with the linguistic refinement, typographical errors, and LaTeX formatting of the manuscript (table generation, figure improvements). The authors retained full control over the scientific content, data analysis, and conclusions.

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A Corpus Construction and Leakage Control

A.1 Merging and Harmonisation

The four corpora are merged into one table with a unified three-way polarity column. VitaminC carries FACT3 labels, which we map onto NLI3 as SUPPORTS to entailment, NOT ENOUGH INFO to neutral and REFUTES to contradiction. SICK is assembled from two public halves, one carrying the inference label and one the relatedness score; the join is verified to lose no more than 1% of rows, and in practice loses none.

A.2 Stratification

Development and test splits take 4 000 rows per class, drawn in equal share from each corpus by water-filling: a corpus smaller than its share is taken whole and the shortfall is redistributed over the larger ones. Without it, SICK falls to 8% of the test split and 0.8% of the development split.

A.3 Leakage

We group rows by source sentence and move whole groups, never rows. VitaminC reuses one piece of evidence across many claims, so 1 662 source sentences, 5.6% of the test groups, were also present in training. We drop 2 379 training rows and 229 development rows, and assert that train/dev, train/test and dev/test share no pair and no source sentence. The assertion runs again after augmentation, since a generated pair inherits its source sentences.

A.4 Augmentation Quotas

The three generated families enter in equal quotas of 25 000, which takes the training set from 152 060 rows to 227 060. Mirroring alone would take the contradiction class from 50 641 to 101 282 against 51 274 entailments.

B Training Details

B.1 Optimisation

Table 9 gives the settings, which are identical for all 140 runs. The library version is part of them: a later major release of `transformers` sends DeBERTaV3-large to a non-finite gradient within a hundred steps, on the same corpus and the same settings, and reports the majority-class floor rather than failing. Nothing is tuned per encoder: one setting across the grid is what makes the seven encoders comparable, and the cost is that no encoder sits at its own optimum.

B.2 Head Reuse

Four of the seven checkpoints ship a three-way inference head. We keep it when its class order matches ours and permute its rows when the published order is reversed, as it is for `roberta-large-mnli`. The alternative, training from a permuted head, is silent and costs an epoch of relearning the permutation.

B.3 Seeds

Seeds [42, . . . , 51] control initialisation of the new head, data order and dropout. Unless stated otherwise, every number reported in this paper is a mean over the ten with its standard deviation.

Setting	Value
Objective	cross-entropy, three classes
Optimiser	AdamW, default β
Learning rate	10^{-5} , linear decay
Epochs	2
Effective batch	32
Micro-batch	8 (base), 4 (large)
Gradient accumulation	4 (base), 8 (large)
Max sequence length	256 tokens
Precision	fp32
Library	transformers 4.57
Seeds	[42, ..., 51]

Table 9: Optimisation settings, identical across the grid. The micro-batch is sized for the smallest card, the accumulation holds the effective batch at 32.

C Hardware and Compute Budget

C.1 Hardware

The grid runs on three NVIDIA Pascal-generation cards: a GTX 1080 Ti (11 GB), a Quadro P5000 (16 GB) and a Titan Xp (12 GB). All training is fp32, since the generation has no usable bfloat16. Two of the seven encoders wedge the GTX 1080 Ti reproducibly, which we handle by excluding those two from that card rather than by diagnosing the driver fault. Encoders built around FlashAttention and unpadded attention need compute capability 8.0 or better; the runner refuses them on this hardware rather than running a degraded version that no comparison could use.

C.2 Budget

The grid takes roughly 500 GPU-hours in total. A base-encoder run takes 68 minutes on the RAW condition and 109 minutes on AUG; a large-encoder run takes 318 and 489 minutes. The composition of Section 7 trains nothing: it is two forward passes over 4 116 pairs plus a grid search over one scalar, and runs on CPU in under half an hour. No private API was used, and the total monetary cost is the electricity of hardware we already own.

D Composition with an Off-the-Shelf Head

The two heads are composed with the same magnitude model at $\alpha = 2$ and scored on the SICK test half. Table 10 gives the trade-off and Figure 4 gives its cause. The off-the-shelf head wins the sign criterion and loses the combined objective, because it is confident on pairs that are merely unrelated: in the bins where it predicts a contradiction probability between 0.4 and 0.9, the observed rate stays near one in ten, so the product subtracts magnitude from pairs that should keep theirs.

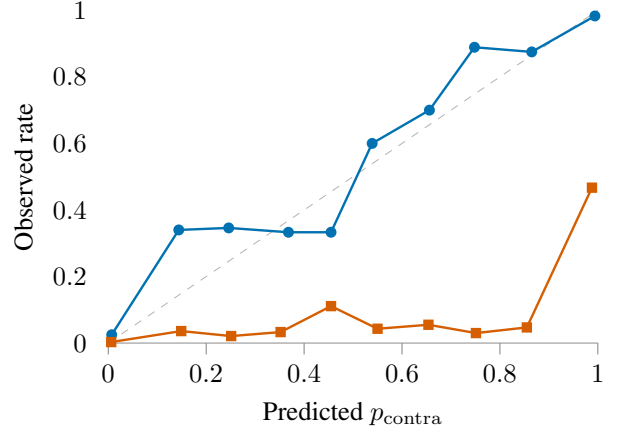


Figure 4: Reliability of p_{contra} on the SICK test half, ten bins. The dashed diagonal is perfect calibration, where a predicted probability equals the observed rate. The **fine-tuned head** tracks it, the **off-the-shelf head** falls far below.

Polarity head	Published	Fine-tuned
Contradictions below zero	97.61	83.71
Relatedness r	0.599	0.795
Combined objective	0.574	0.665

Table 10: One magnitude model composed with two polarity heads at $\alpha = 2$, SICK test half. First row in percent of the 712 contradictions. Higher is better throughout.

The fine-tuned head tracks the diagonal over the same range, which is what the third criterion of Equation 2 rewards.